COMPUTER ENGINEERING

QUEZON CITY

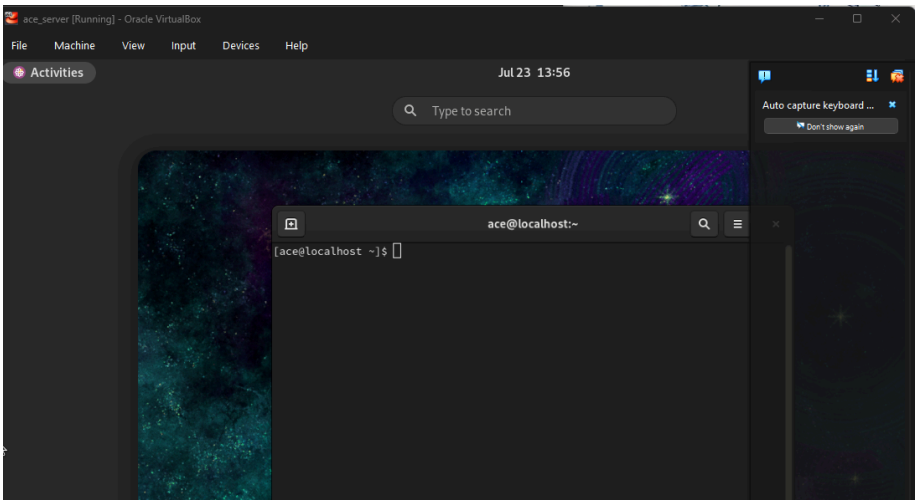
COURSE CODE/TITLE:	CPE212 Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07/23/26	**DATE SUBMITTED:**	07/23/26
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ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

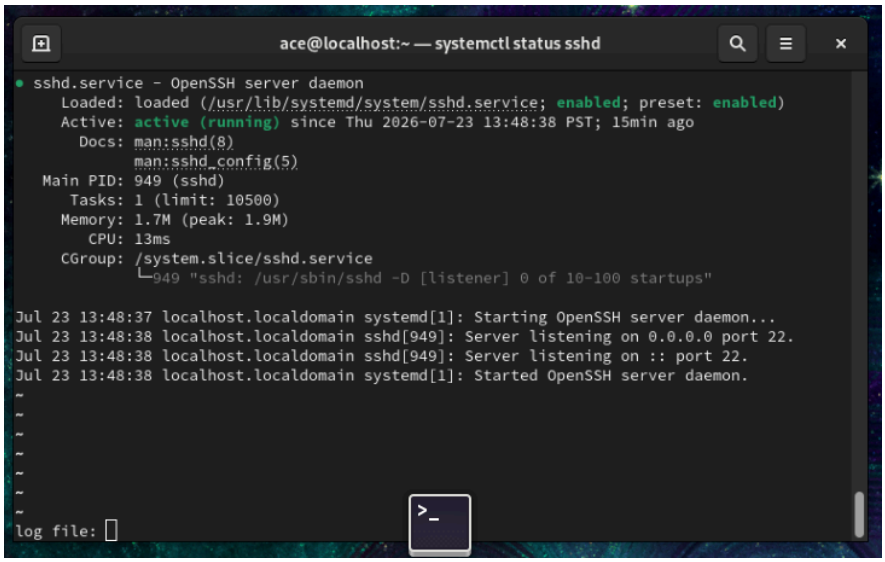
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:
   $ dnf install openssh-server



2. Start the *sshd* daemon and set to start after reboot:
   $ systemctl start sshd
   $ systemctl enable sshd

```
[ace@localhost ~]$ systemctl start sshd
[ace@localhost ~]$ systemctl enable sshd
[ace@localhost ~]$
```

3. Confirm that the sshd daemon is up and running:
   $ systemctl status sshd

```
[ace@localhost ~]$ systemctl start sshd
[ace@localhost ~]$ systemctl enable sshd
[ace@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-07-23 13:48:38 PST; 23min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 949 (sshd)
      Tasks: 1 (limit: 10500)
     Memory: 1.7M (peak: 1.9M)
        CPU: 13ms
     CGroup: /system.slice/sshd.service
             └─949 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 13:48:37 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Jul 23 13:48:38 localhost.localdomain sshd[949]: Server listening on 0.0.0.0 port 22.
Jul 23 13:48:38 localhost.localdomain sshd[949]: Server listening on :: port 22.
Jul 23 13:48:38 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
[ace@localhost ~]$
```

4. Open the SSH port 22 to allow incoming traffic:
   $ firewall-cmd --zone=public --permanent --add-service=ssh
   $ firewall-cmd --reload

```
[ace@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[ace@localhost ~]$ firewall-cmd --reload
success
[ace@localhost ~]$
```

5. Locate the ssh server man config file */etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the */etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:
   $ systemctl reload sshd

```
# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
"/etc/ssh/sshd_config" 130L, 3674B
```

```
[ace@localhost ~]$ sudo sshd -t
[ace@localhost ~]$ sudo systemctl reload sshd
[ace@localhost ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-07-23 13:48:38 PST; 58min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 7865 ExecReload=/bin/kill -HUP $MAINPID (code=exited, status=0/SUCCESS)
   Main PID: 949 (sshd)
      Tasks: 1 (limit: 10500)
     Memory: 1.7M (peak: 2.0M)
        CPU: 27ms
     CGroup: /system.slice/sshd.service
             └─949 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 13:48:37 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Jul 23 13:48:38 localhost.localdomain sshd[949]: Server listening on 0.0.0.0 port 22.
```

Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.
2. Using the command *ssh-copy-id*, connect your local machine to CentOS.
3. On CentOS, verify that you have the *authorized_keys*.

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
   - When choosing between Debian and Red Hat, we should consider what purpose of a system we need. If it's for a small business server that needs guaranteed support and long term stability, Red Hat is usually the better choice because it offers official paid support and is widely used in corporate environments. If we want something more flexible, and have community support, Debian is a great option, it is ideal for personal projects or general purpose servers.

2. What are the main differences between Debian and Red Hat Linux distributions?
   - The main difference is their package management systems. Debian uses the DEB package format with APT, while Red Hat uses YUM/DNF. Debian is a free, community driven distro that many other distros like Ubuntu are built on, while Red Hat is commercially developed, with CentOS traditionally being a free rebuild of RHEL. Red Hat also offers official paid enterprise support, whereas Debian relies only on community support and is completely free.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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